

Daily Report

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1 Presentation and Research Review

Yesterday, I prepared a presentation on my review and what I learned from forecasting models, specifically Reservoir Computing (RC) and ECG signal filtering models. I presented it and discussed my research insights.

1.1 Insight: LoRA-style Adaptation for Reservoir Computing

I noted that Reservoir Computing is conceptually similar to the LoRA (Low-Rank Adaptation) technique, and raised the question of why LoRA has not been applied to Reservoir Computing in order to shift the random or predefined parameters toward better results.

Interestingly, a paper published only about three months ago proposed a closely related idea [4]. The paper shows that low-rank adapters trained over a **frozen random backbone** can recover 96–100% of fully trained performance while training only 0.5–40% of the parameters.

While this paper is not specific to Reservoir Computing or Echo State Networks, I believe the underlying idea is closely related: both approaches keep a large set of parameters fixed (random) and train only a small, low-rank component on top. This suggests a promising direction, applying a LoRA-style adapter to Reservoir Computing to shift the random reservoir parameters toward more meaningful, task-relevant dynamics.

Proposed Direction

I suggested that we test this idea on our own task, to evaluate whether combining LoRA with Reservoir Computing can achieve state-of-the-art results with significantly lower computational cost. I believe this could be especially beneficial for Reservoir Computing specifically, since, similar to Echo State Networks, the reservoir consists of random neurons. Introducing a small trained component could shift these random dynamics toward something more meaningful, allowing us to benefit from both:

- A very small trainable component (low computational cost), and
- The inherent non-linearity and high-dimensional projection properties of the reservoir.

1.2 Additional Idea: RC as a Feature Extractor

I also suggested that Reservoir Computing could be used as a **feature extractor** placed before any sequential model. Since RC is fast and computationally cheap, it could offload part of the temporal feature extraction burden from recurrent models such as LSTMs, potentially simplifying the downstream model and reducing training time.

2 ECG Dataset Exploration and Preprocessing

We had a task to explore the ECG dataset and begin preprocessing and denoising it. I read three papers (see References section) and decided to try four preprocessing pipelines:

1. **Wavelet only (SWT)**

From: “*Comparative evaluation of filtration techniques for ECG signal denoising with emphasis on stationary wavelet transform*” [1]

2. **Notch + Wavelet (SWT)**

My own suggestion after reading all the papers, to remove PLI (power-line interference) noise before applying SWT

3. **Median + Bandpass filter**

From: *FADE: Forecasting for Anomaly Detection on ECG* [2]

Median filter (kernel size = 3) + Bandpass filter (0.5 to 30 Hz)

4. **SMA + Median + Notch + Normalize**

From: *ECG Forecasting System Based on Long Short-Term Memory* [3]

SMA (window = 3) + Median (window = 3) + Notch (60 Hz, order 2) + Normalize [0, 1]

3 Future Work (Weekend): Continuing the Preprocessing Pipeline

- **Normalization strategy**

Compare how each paper normalizes signals (z-score, min-max [0,1], empirical clamping + range normalization) and decide which fits our forecasting task best. Some papers avoid normalization when the data contains abnormalities (to preserve the raw abnormal characteristics), while others do normalize. I need to spend more time researching this to decide the right approach for our case.

- **Train/test split strategy**

- **How to compare preprocessing pipelines using baseline forecasting models**

References

- [1] Ádám, N., Val’ko, D., Balogh, Z., Madoš, B., & Hurtuk, J. (2025). *Comparative evaluation of filtration techniques for ECG signal denoising with emphasis on stationary wavelet transform*. Scientific Reports, 15, 42514. <https://doi.org/10.1038/s41598-025-26476-1>
- [2] Ruiz-Barroso, P., Castro, F. M., Miranda, J., Constantinescu, D-A., Atienza, D., & Guil, N. (2025). *FADE: Forecasting for Anomaly Detection on ECG*. arXiv:2502.07389.
- [3] Zacarias, H., Lôbo Marques, J. A., Felizardo, V., Pourvahab, M., & Garcia, N. M. (2024). *ECG Forecasting System Based on Long Short-Term Memory*. Bioengineering, 11(1), 89. <https://doi.org/10.3390/bioengineering11010089>
- [4] Hazan, H., Zhang, Y., Hartl, B., & Levin, M. (2026). *A Little Rank Goes a Long Way: Random Scaffolds with LoRA Adapters Are All You Need*. arXiv:2604.08749. <https://arxiv.org/abs/2604.08749>

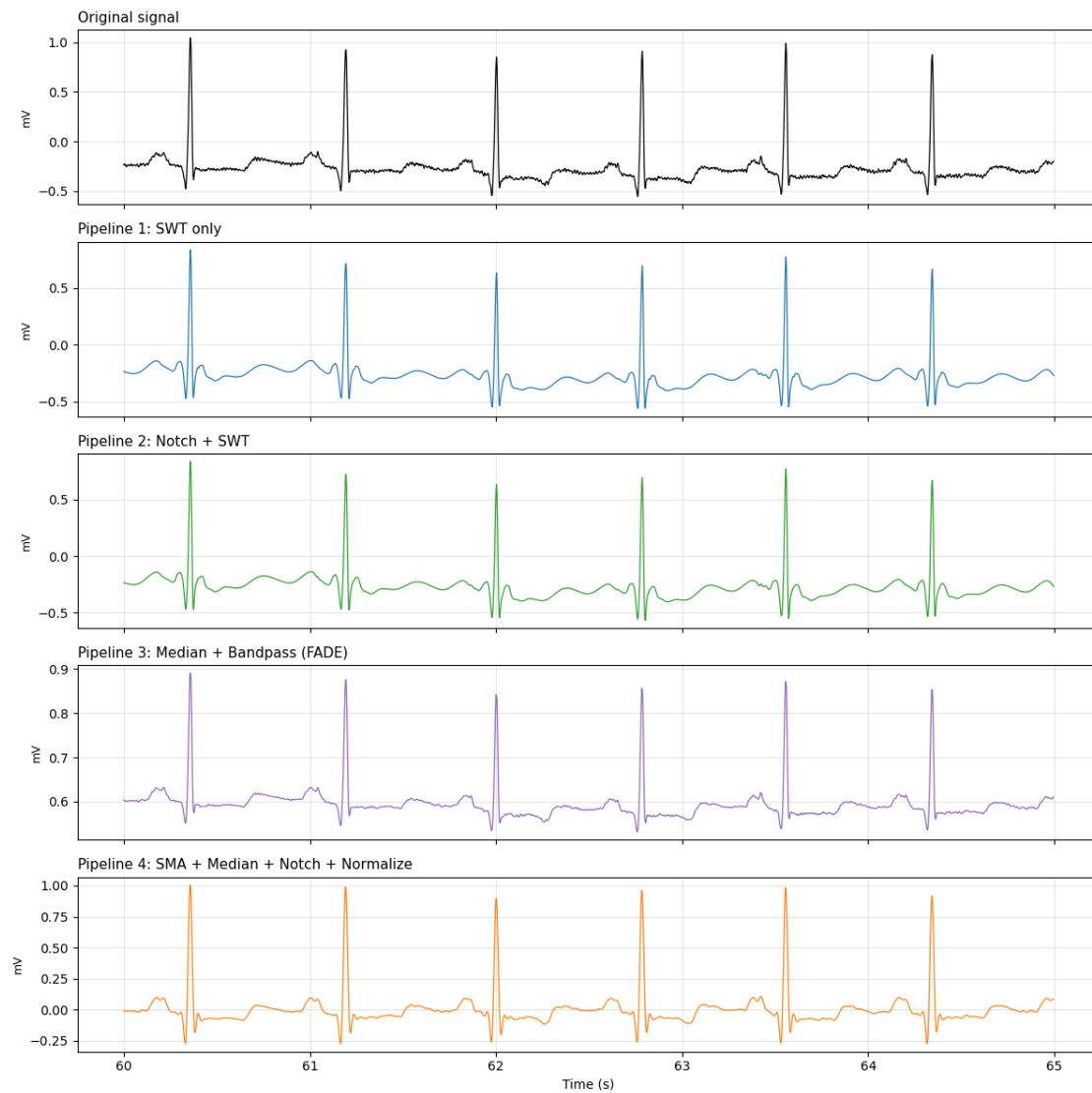


Figure 1: Preprocessing results comparison across the four pipelines.